

Amazon ML Challenge 2026 — Business Entity Resolution

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1. Problem Overview

Business Entity Resolution

You are given business records coming from **three independent data sources**. These sources contain records describing real-world businesses, but there is **no common identifier/shared ID** that directly tells you which records belong to the same business.

The records can contain different representations of the same business. For example, the same business may appear with:

- Different forms of its name
- Different abbreviations or legal suffixes
- Slightly different address formats
- Different address descriptions or landmarks
- Missing or rearranged address components
- Other textual variations

Your task is to determine **which records from Source 2 and Source 3 correspond to each business in Source 1**.

Source 1 acts as the reference/anchor set.

For every Source 1 entity, you must identify **all matching records in Source 2 and/or Source 3**.

A Source 1 entity can therefore have:

- **No match**
- **One match**
- **Multiple matches**

This is a large-scale **Entity Resolution / Record Linkage** problem. The published challenge materials describe the same Source 1 → Source 2/Source 3 matching setup.

2. What the Problem Looks Like

The video gives a concrete example using **Acme Robotics**.

A business signs up with:

Business name:

Acme Robotics Inc

Business address:

500 Market St, San Jose, CA

The challenge specifically says that, although phone and email may exist in the underlying real-world data, **this challenge uses only the business name and address** in this example.

The same business may subsequently appear in other sources as:

Source 1

Name: Acme Robotics Inc.

Address: 500 Market St, San Jose

Source 2

Name: Acme Robotics Incorporated

Address: 500 Market Street, San Jose CA

Source 3

Name: Acme Robotics

Address: Nr. City Hall, San Jose

These are three different records, with different textual representations, but they refer to **the same real-world business**.

The key challenge is therefore:

Given several noisy records without a shared identifier, determine which records refer to the same underlying real-world entity.

3. Expected Relationship

Conceptually, you are building links of the form:

Source 1 entity

```
|
+---- matching Source 2 record
|
+---- matching Source 3 record
```

But the number of links is not fixed.

For example:

```
S1-Acme
├── S2-Acme
└── S3-Acme
```

is valid.

But another Source 1 entity could have:

```
S1-X
└── nothing
```

and another could have:

```
S1-Y
├── S2-Y1
├── S2-Y2
├── S3-Y1
└── S3-Y2
```

So this is **not a simple one-to-one matching problem**.

4. Blocking / Candidate Generation

The video introduces an important first stage called **blocking**.

The fundamental problem is that comparing every Source 1 record with every Source 2 and Source 3 record would create an enormous number of pairwise comparisons.

Therefore, instead of comparing everything against everything, you first construct a smaller set of **plausible candidates**.

Blocking

The video illustrates blocking using a cheap key based on:

name + address

Records are placed into blocks when their names and/or addresses suggest that they could correspond to the same entity.

The video explicitly summarizes the idea as:

Blocking buckets records by a cheap key built from name and address, so similar names or shared addresses group together, matches and look-alikes alike.

Important implication

A block can contain:

- Genuine matches
- False candidates
- Look-alikes
- Records that happen to share an address
- Records with similar names but different businesses

Therefore, **blocking does not perform the final matching**.

It only generates candidates that the actual matching system must subsequently evaluate.

5. Blocking Example Shown in the Video

The video gives the following small example.

Source 1

ID	Business	Address
S1-732914	Acme Robotics Inc	500 Market St, San Jose
S1-889301	Delta Foods	8 Oak Ave, Austin
S1-410562	Bright Cafe LLC	22 Pine St, Reno
S1-205774	Zen Traders	4 Hill Rd, Boise

Source 2

ID	Business	Address
S2-118820	Acme Robotics Inc	500 Market Street, San Jose
S2-540221	Acme Robotix	12 Elm Rd, San Jose
S2-397155	Delta Foods Co	8 Oak Avenue, Austin
S2-663049	Bright Cafe	22 Pine Street, Reno

Source 3

ID	Business	Address
S3-905477	Acme Robotics	Nr City Hall, San Jose
S3-063118	Acme Bakery	500 Market St, San Jose
S3-651230	Delta Foods Ltd	Oak Ave, Austin
S3-472088	Kappa Motors	90 Lake Dr, Fargo

For the Source 1 entity:

S1-732914

the video shows the generated **Acme block** containing:

S2-118820
S2-540221
S3-905477
S3-063118

This is represented in:

candidate_pairs.tsv

So the blocking stage produces a **shortlist of candidates**, and the later matching stage has to determine which of those candidates are actual matches.

This example also demonstrates why blocking alone is insufficient:

- S2-118820 is a very plausible match.
- S3-905477 is also a plausible match despite the different address representation.
- S2-540221 is a similar-looking name but has a different address.
- S3-063118 shares the address but represents **Acme Bakery**, not Acme Robotics.

The model therefore needs to distinguish **true matches from look-alikes**.

6. Dataset

The challenge provides two major data situations: **training** and **test**.

Training Set

The training data contains records from **all three sources**.

You also receive ground-truth labels in:

train_ground_truth.tsv

These labels allow you to:

- Develop your entity-resolution system
- Train the matching model
- Evaluate your approach locally

The ground truth is organized as **one row per Source 1 entity**.

Conceptually:

source1_entity_id	matched_entity_ids
S1-123	S2-456, S3-789
S1-124	S2-999
S1-125	

An empty list means that the Source 1 entity has **no corresponding Source 2/3 records**.

All challenge files are **tab-separated .tsv files**, so they should be parsed using tab separation rather than comma separation.

7. Test Set

The test set contains the same three-source structure:

Source 1

Source 2

Source 3

but **without ground-truth labels**.

You must produce predictions for **every Source 1 entity**.

The test predictions are what the leaderboard evaluates.

So the overall workflow is:

Training data

↓

Develop / train / validate model

↓

Apply model to test data

↓

Predict matches for every Source 1 entity

↓

Submit predictions

8. What You Actually Need to Predict

For every Source 1 entity, output the IDs of **all Source 2 and Source 3 records that represent the same real-world business**.

For example:

S1-732914 S2-118820, S3-905477

means:

Source 1 entity S1-732914 matches Source 2 record S2-118820 and Source 3 record S3-905477.

For an entity with no corresponding records:

S1-205774

with an empty match list means:

There is no matching Source 2 or Source 3 record.

9. Submission During the Challenge

During the challenge, you submit:

`matching_results.tsv`

This is the file that is scored on the leaderboard.

The requirements shown in the video are:

- One row for every Source 1 entity
- The row contains the predicted matching Source 2/3 IDs
- An empty list means no match

The published challenge specification likewise identifies `matching_results.tsv` as the leaderboard-scored prediction file.

10. Final Submission Package

At the end, the challenge requires a **.zip package**.

The video specifies that it should contain:

output/matching_results.tsv

The final predicted matches.

output/candidate_pairs.tsv

The blocking/candidate set generated by your system.

This is specifically included so that the candidate-generation process can be **audited**.

code/...

A runnable implementation of your pipeline.

Documentation_template.md

A description of your methodology.

The published challenge materials also specify these same core final-package components.

11. Validation Before Submission

The video recommends running:

`utils/validate_submission.py`

before submission.

Its purpose is to detect formatting or structural problems before you submit.

This matters because a correct ML prediction can still fail if the output file is malformed.

12. Evaluation Metric

The competition uses:

Macro $F_{0.5}$

The formula shown in the video is:

$$F_{0.5} = \frac{1.25 \times Precision \times Recall}{0.25 \times Precision + Recall}$$

This corresponds to the general F_{β} score with:

$$\beta = 0.5$$

Because $\beta < 1$, **precision is given more weight than recall.**

13. What That Means for Your Model

The video makes a very important observation:

A false merge costs about twice as much as a miss.

In practical terms:

False merge

You say:

These two records are the same business.

when they actually represent two different businesses.

This is heavily penalized.

Miss

You fail to identify a real match.

This is also bad, but the metric places relatively greater emphasis on avoiding incorrect merges.

Therefore, the challenge is intentionally **precision-heavy**.

The video summarizes the practical implication as:

When unsure, do not merge.

The published challenge specification describes the same precision-heavy $F_{0.5}$ behavior.

14. Singleton Entities

A particularly important special case is a **singleton**.

A singleton is a Source 1 entity for which:

there is no matching record in Source 2 or Source 3

Therefore, the correct prediction is:

empty list

The video explicitly emphasizes:

- Correctly predicting a singleton as empty earns a full **1.0**
- Predicting any match for a true singleton gives **0**

So singleton detection is an important component of the problem rather than a minor edge case.

15. Macro Averaging

The metric is **macro F_{0.5}**, meaning the score is considered at the Source 1 entity level and then aggregated across Source 1 entities rather than simply pooling all individual matches together. The published specification makes this per-Source-1 averaging explicit.

This makes performance on individual Source 1 businesses important.

16. Blocking Creates a Recall Ceiling

One of the most important conceptual points in the video is:

Blocking sets your recall ceiling.

Why?

Suppose the real match is:

S1-A → S3-B

but your blocking system never places S3-B into the candidate set for S1-A.

Then the matching model never even gets the opportunity to consider that pair.

Therefore:

Bad blocking

↓

True match never considered

↓

Matching model cannot recover it

↓

False negative

Hence the video advises:

Invest in blocking first.
This is a fundamental architectural constraint of the challenge.

17. Regional Variation

The video also specifically warns contestants to:

Account for region-specific patterns in both names and addresses.

In other words, the way businesses and addresses are represented may depend on the geographical region.

A robust entity-resolution system therefore should not assume that:

same formatting conventions

+

same abbreviations

+

same naming patterns

apply universally.

Your normalization, blocking and matching strategy needs to be capable of handling geographical variation.

18. No External Data Lookup

This is an explicit challenge restriction.

The video states:

No external data lookup.

Specifically, the video says that:

- External databases are prohibited
- External APIs are prohibited
- Geocoding services are prohibited
- Only the provided challenge data may be used

So you **cannot** solve the problem by taking an address or business name and querying an external service to determine the real business.

The model must perform entity resolution using the information supplied by the competition. The published challenge materials state the same restriction.

19. Complete End-to-End Problem Definition

Putting everything together, the challenge can be expressed mathematically as follows.

You are given:

$$S_1 = \{r_1, \dots, r_N\}$$

$$S_2 = \{r'_1, \dots, r'_M\}$$

$$S_3 = \{r''_1, \dots, r''_K\}$$

where each record contains noisy business information, primarily including the business identity information supplied by the challenge.

There is no shared identifier across the three sources.

For every Source 1 record r_i , you must predict a set:

$$\hat{M}_i \subseteq S_2 \cup S_3$$

such that $\hat{M}_i$ contains the Source 2/3 records that refer to the same underlying real-world business as r_i .

The prediction may be:

$$\hat{M}_i = \emptyset$$

or contain one record, or contain multiple records.

The overall system therefore needs to solve two closely connected problems:

Stage 1 — Candidate Generation / Blocking

Reduce the enormous possible pair space to a manageable set of plausible candidate pairs.

$$S_1 \times (S_2 \cup S_3) \longrightarrow C$$

where:

$$C \subseteq S_1 \times (S_2 \cup S_3)$$

The critical requirement is to preserve true matches.

Stage 2 — Matching

For every candidate pair:

$$(r_i, r_j) \in C$$

determine whether:

$$r_i \equiv r_j$$

in the sense that both records represent the same real-world business.

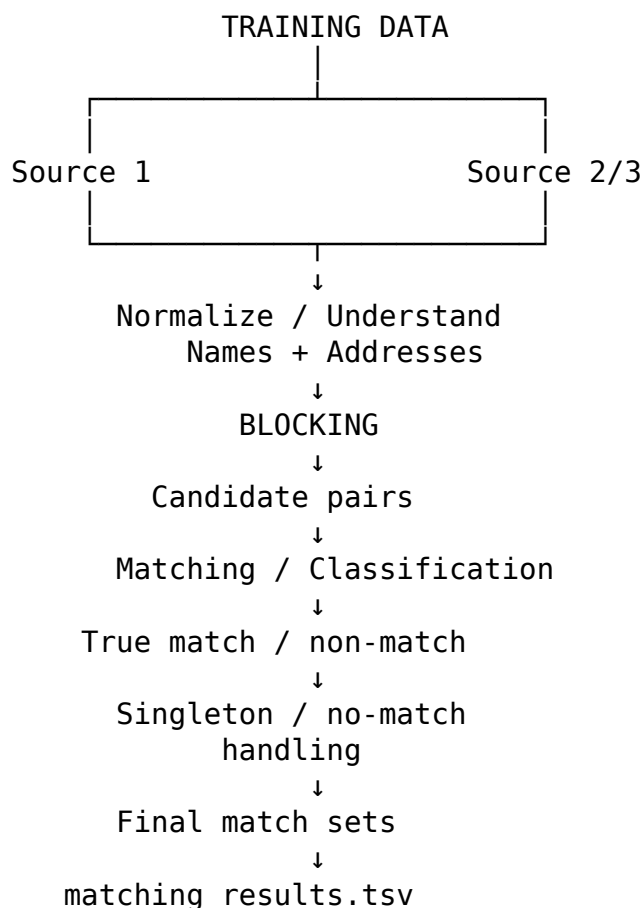
Then generate the final match set for every Source 1 entity.

20. The Challenge in One Sentence

Build an ML-based entity-resolution system that takes three independent collections of noisy business records with no shared identifiers, identifies all Source 2 and Source 3 records corresponding to each Source 1 business—including the possibility of no match or multiple matches—and produces predictions optimized for macro $F_{0.5}$, where precision is weighted more heavily than recall, while using only the provided challenge data.

21. What the Video Is Really Asking You to Build

The intended pipeline can therefore be visualized as:



The central difficulty is **not simply string matching**.

You need to simultaneously deal with:

no shared IDs + noisy names + noisy addresses + false candidates + multiple matches + zero matches + singleton entities + recall limitations caused

by blocking + precision-heavy evaluation.